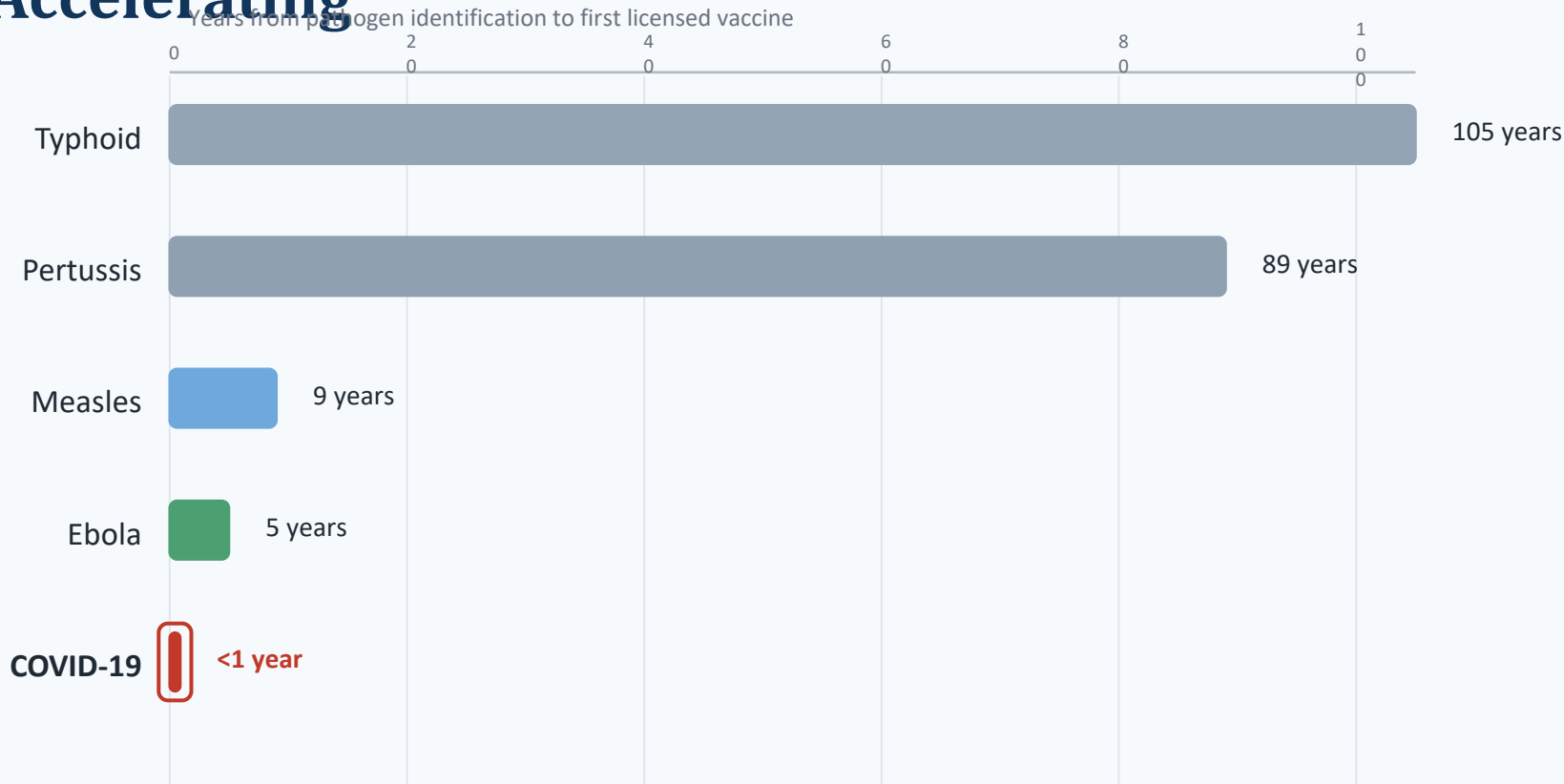


The Power of Vaccination:

From Smallpox Eradication to COVID-19 in Record Time

1796–2020 | Data-driven evidence for policy and public health action

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



Key point: Development timelines collapsed from over a century (typhoid: 105 years) to under one year (COVID-19).

Source figures and durations from source_brief.md (Our World in Data vaccination innovation reference).

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

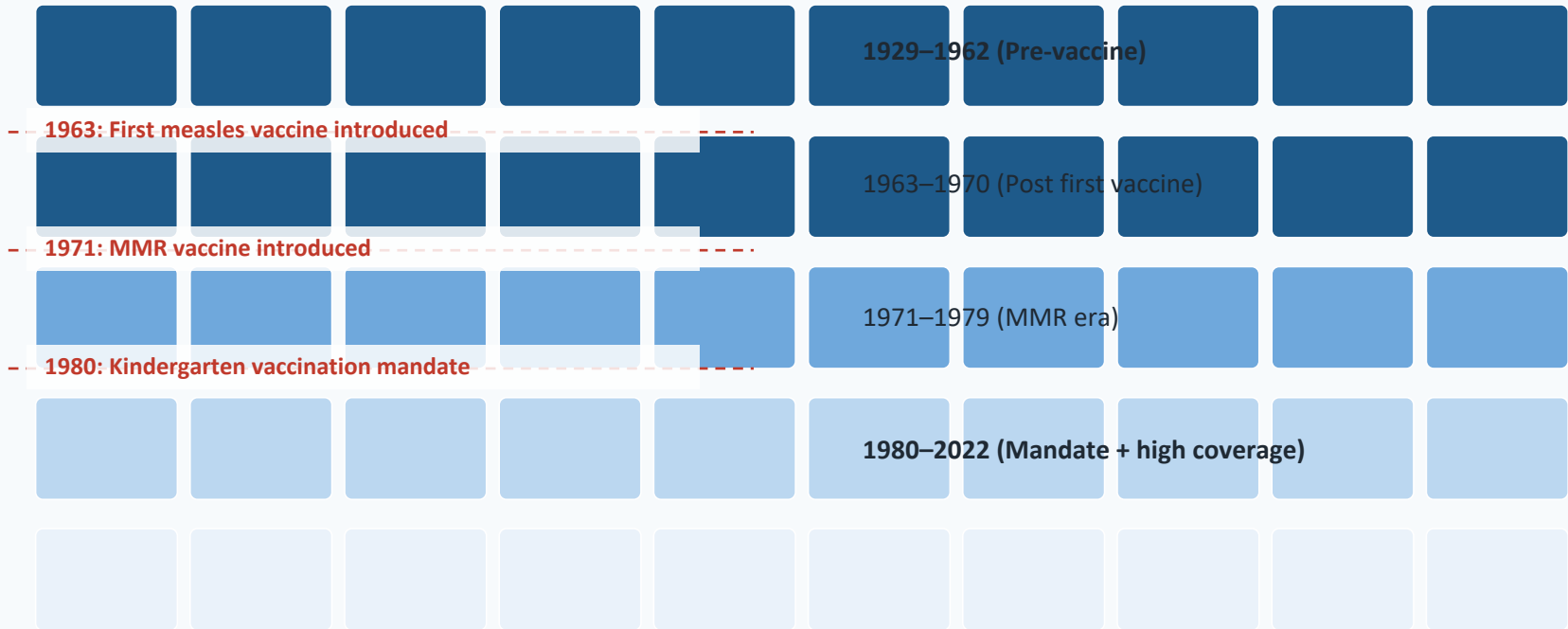
Disease	Pre-vaccine annual burden	Current annual burden	Case reduction	Death reduction	Status
Diphtheria	High historical burden	0	100%	100%	Eliminate
Polio (paralytic)	High historical burden	0	100%	100%	Eliminate
Smallpox	High historical burden	0	100%	100%	Eradicated
Hepatitis A	—	—	89%	—	Reduced, need
Acute Hepatitis B	—	—	83.9%	—	Reduced, not major
Pneumococcal disease	—	—	40.5%	31.3%	gap remains

Diseases with 100% reduction are marked in green. Persistent burden remains for pneumococcal disease and viral hepatitis.

All percentages and reduction claims are taken directly from [source_brief.md](#).

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Illustrative heatmap-style panel (state incidence categories by period)



Pattern shown in source: widespread high incidence before 1963, then near-zero levels after 1980 with only sporadic outbreaks.

Milestone years (1963, 1971, 1980) and narrative drawn from source_brief.md measles heatmap reference.

Smallpox: The Only Human Disease Eradicated Through Vaccination

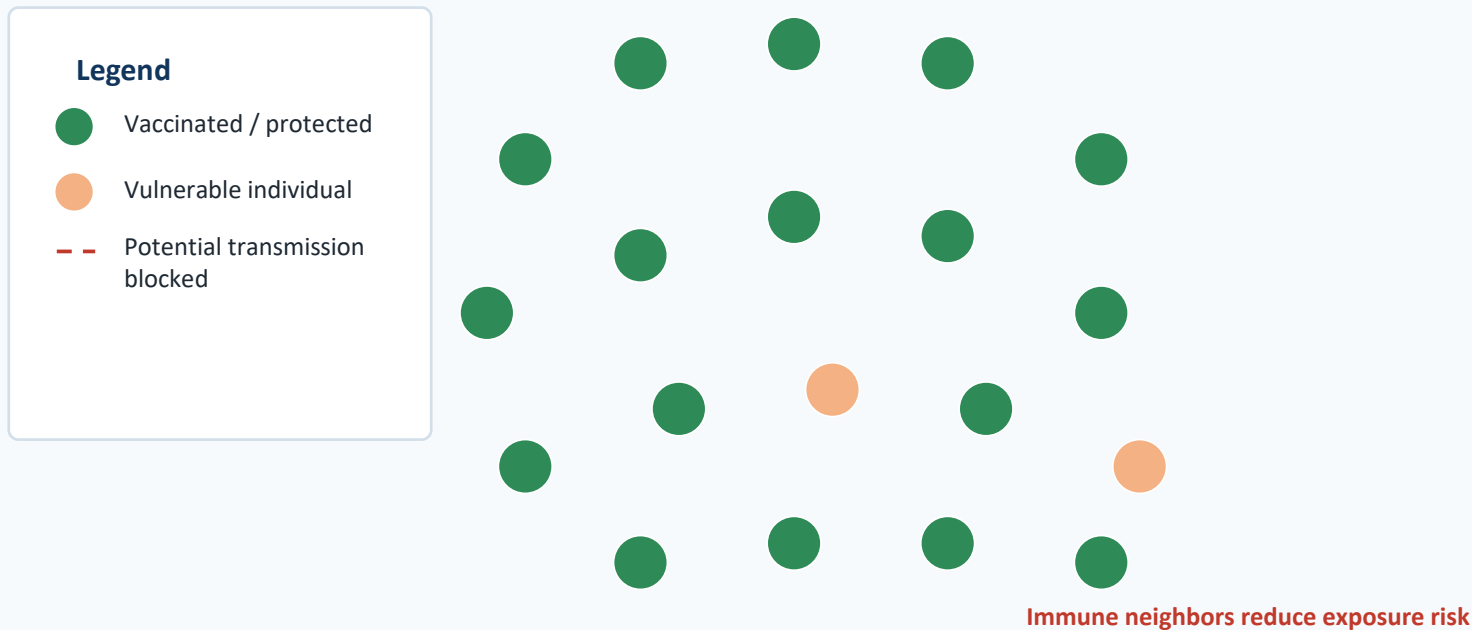


Why eradication succeeded

- Virus infected only humans
- Distinctive symptoms enabled rapid case detection
- Vaccine was effective, cheap, and scalable

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

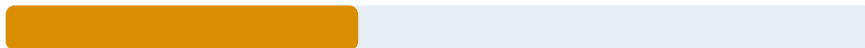
When enough people are vaccinated, transmission chains break — indirectly protecting those who cannot be vaccinated.



Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Reduction from pre-vaccine burden (%)

Pneumococcal disease



40.5%

31.3% death reduction

Hepatitis A



89%

Acute Hepatitis B



83.9%

Barriers to full impact

- Unequal global vaccine access
- Supply shortages and delivery bottlenecks
- Low trust and misinformation
- Need for newer or broader vaccines

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- Vaccines delivered 100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.
- Development timelines compressed from 105 years (typhoid) to under one year (COVID-19).
- Measles incidence fell from widespread high rates to near zero across all states after vaccine-era policy milestones.
- Smallpox eradication in 1980 remains the strongest proof that coordinated vaccination can end a disease globally.
- Coverage and impact gaps persist for pneumococcal disease and hepatitis A/B.

Call to action: Strengthen global access, close coverage gaps, and invest in confidence-building to sustain vaccination gains.